SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	**Total Marks:**	100 marks

CO2 Statement: Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.

Student Name:	U.Divya	**Reg. No.:**	192372277
Section / Batch:	CSE-AI / 4th year	**Date:**	21/10/26

Code of Conduct

I U.Divya/192372277 (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: U.Divya

Instructions

- This test contains FIVE questions.
- Duration: 60 minutes (homework)

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	

Date _____

Annexure A - QUESTIONS

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

Deliverables:

- Figma (.fig) design
- GitHub repository containing exported images/PDF and documentation (README).

1. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.

Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.

Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.

1. Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.

Deliverables:

- Source code/design files.
- GitHub repository with implementation, screenshots, and analysis report.

1. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks.

Analyze:

- How Mobile IP maintains uninterrupted communication.
- Challenges during handoff between networks.
- Security vulnerabilities during mobility.
- Possible design improvements to increase reliability.

Provide navigation between screens demonstrating different mobility scenarios.

Deliverables:

- Interactive prototype.
- GitHub repository containing design files, exported prototype, and documentation.

1. Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.

The dashboard should analyze multiple parameters such as:

- Node mobility.
- Infrastructure availability.
- Network size.
- Routing complexity.
- Energy consumption.
- Reliability requirements.

The system should justify its recommendation based on the analyzed parameters.

Deliverables:

- Source code or Figma project.

GitHub repository with complete code/design, README, screenshots, and analysis report.

Criterion	Marks
Design Quality (Figma/Programming Tool)	20
Technical Analysis	30
Correctness of Mobile Networking Concepts	20
GitHub Repository Organization & Documentation	15
Creativity and Visualization	15
Total	**100 Marks**

Mobile IP Communication Workflow

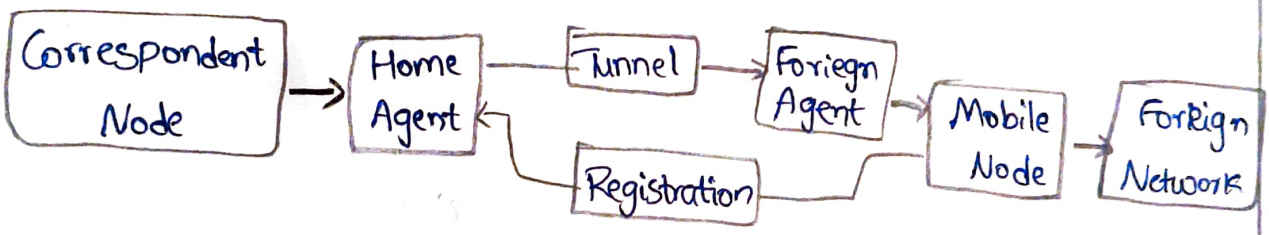
Aim:-

To design a Figma prototype showing Mobile IP Communication when a Mobile Node moves from Home Network to Foreign Network

Procedure:-

1. Open Figma and create a new design
2. Add MN, HA, FA, CN and CoA
3. Draw registration and Packet tunneling arrows.
4. Show mobility, delay and Packet loss
5. Create Prototype navigation and export the design.

Sample Output:-



Result:-

Designed a Mobile IP workflow showing registration, mobility. and Packet tunneling

Cellular Network and MANET Comparison

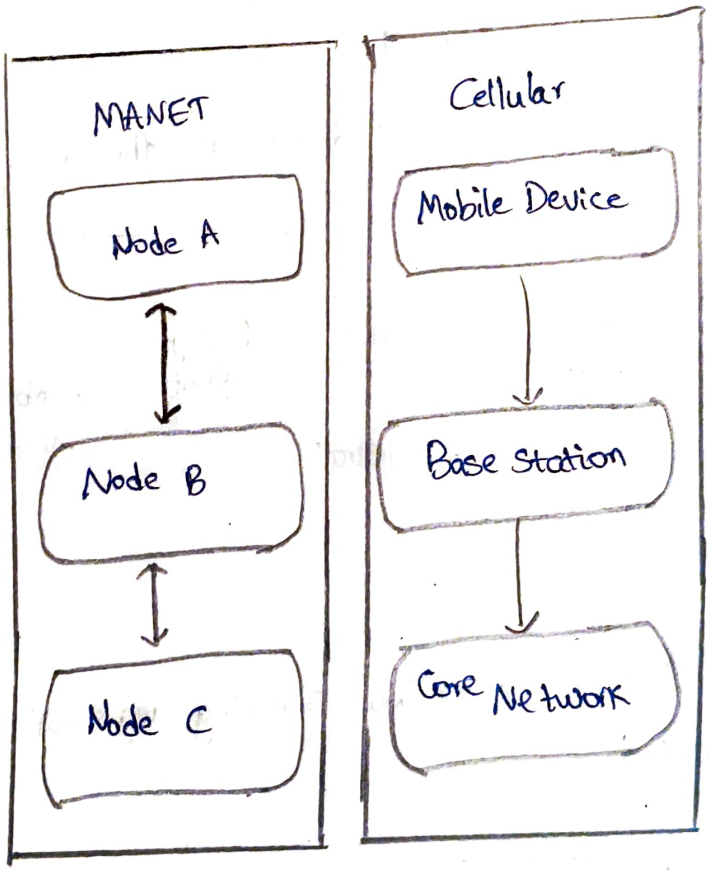
Aim:-

To Compare Cellular Wireless Network and MANET architectures using Figma.

Procedure:

1. Open Figma
2. Create two architecture diagrams.
3. Add base stations and mobile nodes.
4. Show cellular and multi-hop communication
5. Add advantages, limitations and comparison.

Sample Output:-



Result:-

Compared Cellular Networks and MANET based on architectu

and Communication methods.

Routing Protocol Visualization.

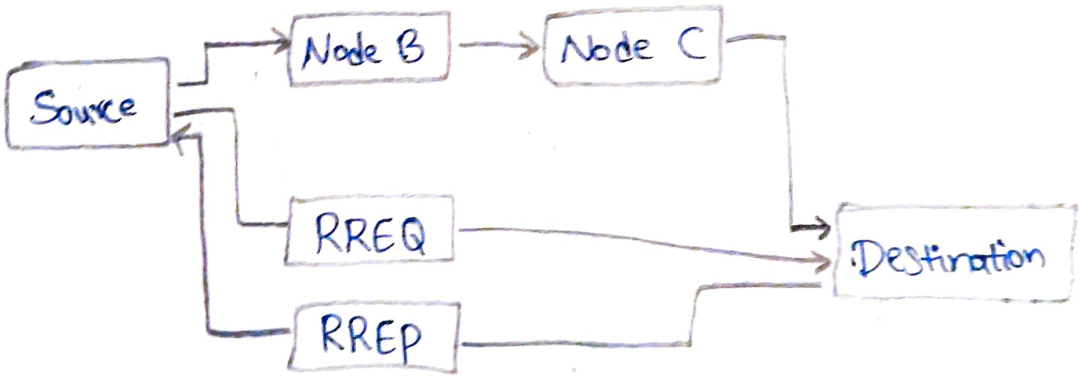
Aim:-

To visualize and compare Proactive DSDV and reactive AODV routing protocols.

Procedure:-

1. Create a wireless network diagram
2. Show DSDV routing tables and periodic updates
3. Show AODV route request and reply.
4. Demonstrate route failure and maintenance
5. Compare delay, overhead and mobility.

Sample Output:-



Result:-

Visualized and compared proactive and reactive routing Protocols

4. Smart Healthcare Mobile IP

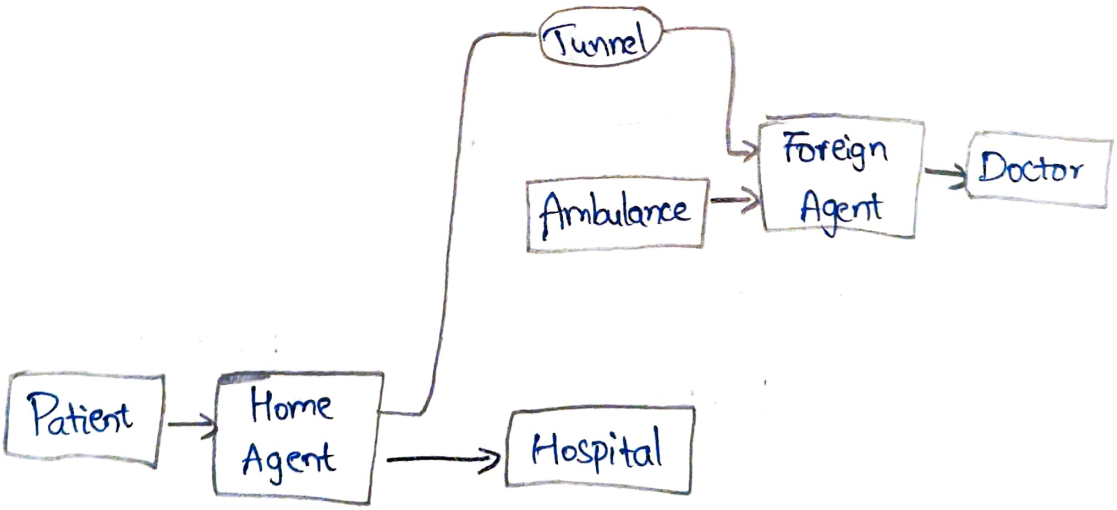
Aim:-

To design a Smart Healthcare IP application for continuous Communication between doctors, patients, ambulances and hospitals

Procedure:-

1. Create healthcare application screens in Figma
2. Add patient, doctor, ambulance and hospital
3. Include HA, FA and CoA
4. Show network handoff and packet tunneling
5. Add Security and navigation features.

Sample Output:-



Result:-

Designed a healthcare prototype demonstrating Mobile IP mobility and communication.

Wireless Network Decision Dashboard

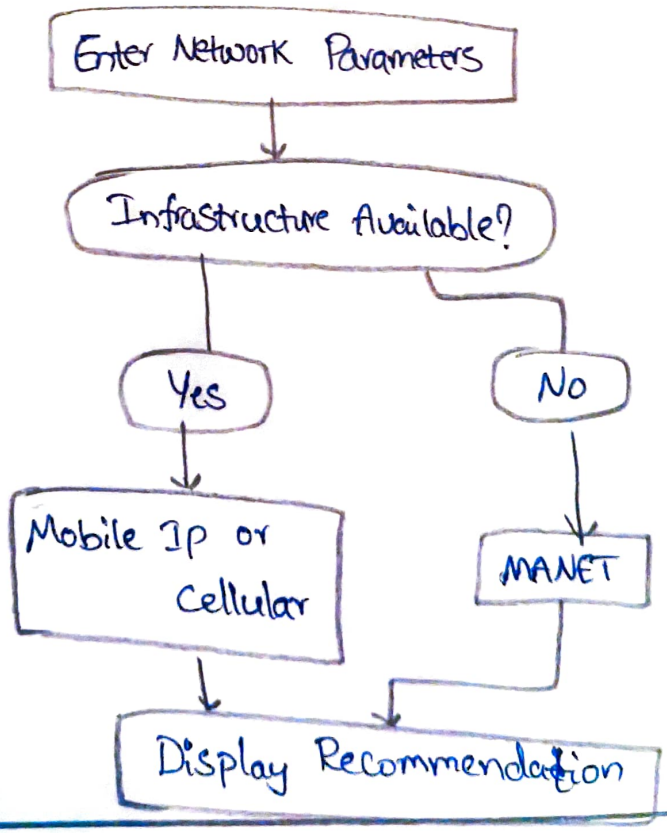
Aim:-

To develop a dashboard that recommends Mobile IP, Cellular Network or MANET based on network requirements

Procedure:-

1. Create a dashboard using HTML, CSS, JavaScript or Figma
2. Add mobility, infrastructure, network size, energy and reliability inputs.
3. Implement decision logic
4. Display the recommended network
5. Add advantages and limitations.

Sample Output:-



Output:-

Designed a dashboard to recommend suitable wireless network architectures.